CURIE, MARIE

THE PHYSICIST MARIE CURIE was a pioneer in the science of radioactivity – the study of powerful rays emitted by certain rare materials. Her work changed physics and chemistry and formed a basis for later research in nuclear physics. She discovered two previously unknown elements and also founded an important research institute. In 1903, she shared the Nobel prize for physics with her husband Pierre, with whom she worked, and French scientist Henri Becquerel. In 1911, she was awarded the Nobel prize for chemistry. She died after suffering for years from an illness caused by exposure to radiation.

Early life

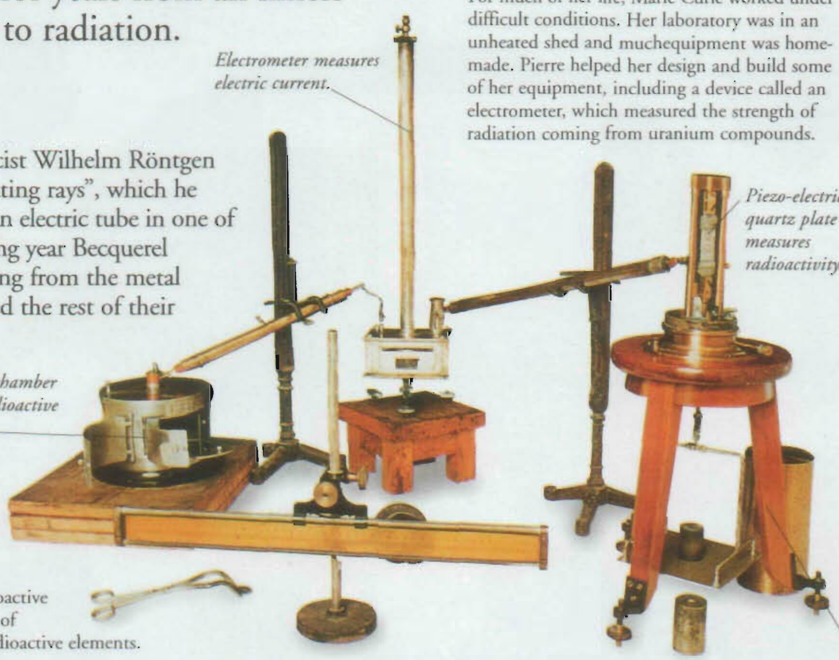
Marie Curie was born in Warsaw, Poland. After finishing school, she worked as a governess to save up money to go to university in Paris. In 1891, she left to study at the Sorbonne. She was top of her year there, in spite of being so hungry sometimes that she fainted during classes.

Poland

At the time that Marie was born, Poland was under the rule of neighbouring Russia, and the best jobs and education went to Russians. After she had finished her schooling, Marie began to go to secret meetings of the "Floating University", a group of Polish people who met to read books that the Russians banned in case they might stir up rebellious ideas.

Equipment

For much of her life, Marie Curie worked under difficult conditions. Her laboratory was in an unheated shed and muchequipment was home-made. Pierre helped her design and build some of her equipment, including a device called an electrometer, which measured the strength of radiation coming from uranium compounds.



Radioactivity

In 1895, the German physicist Wilhelm Röntgen discovered invisible "penetrating rays", which he called X-rays, coming from an electric tube in one of his experiments. The following year Becquerel discovered similar rays coming from the metal uranium. The Curies devoted the rest of their lives to studying these rays.

Isolating radiation

The Curies noticed that pitchblende, the ore from which uranium is extracted, was many times more radioactive than uranium itself. They realized that pitchblende must contain other radioactive substances, so they processed tonnes of pitchblende to extract these other radioactive elements.

Pierre Curie

Born in Paris and educated by his father, Pierre Curie (1859–1906) began work as a laboratory assistant in Paris. He made several important discoveries before he met Marie in 1894. After they married he spent the rest of his life working with her. He made some of her equipment and worked beside her in the laboratory. He was killed in a street accident in 1906.

Pitchblende

The Curies spent about 12 years separating out the radioactive elements in pitchblende. They found there were two substances. One they named polonium, after Marie's home country, and the other they called radium.

X-rays

When World War I (1914–18) broke out, Marie Curie raised funds to set up mobile X-ray units to be used on the battle front. She supervised the conversion of around 200 vans for this purpose. These became known as "Little Curies".

Radium Institute

In 1912, the Sorbonne and the Pasteur Institute decided to found a Radium Institute in Paris, devoted to research into radiation and the medical uses of radioactivity. Marie became a director of the Institute and spent much of her time supporting scientists in their work and raising money for research.

The Joliots

Marie Curie's daughter, Irène, was also a scientist. She and her husband, Frédéric Joliot, worked together, much like Marie and Pierre. They discovered how to make non-radioactive substances radioactive, by bombarding them with radioactive rays. In 1935 they were awarded the Nobel prize for chemistry.

MARIE CURIE

- **1867** Born Manya Skłodowska in Warsaw, Poland
- **1891** Goes to the Sorbonne, Paris; changes first name to Marie
- **1895** Marries Pierre Curie
- **1898** Discovers the elements polonium and radium
- **1903** Awarded Nobel prize for physics
- **1910** After 12 years of work on pitchblende, she produces pure radium for the first time
- **1911** Awarded Nobel prize for chemistry
- **1918** Radium Institute opens after delay owing to World War I; Marie becomes research director
- **1934** Dies in France

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